Imperial College
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Chip Design projects

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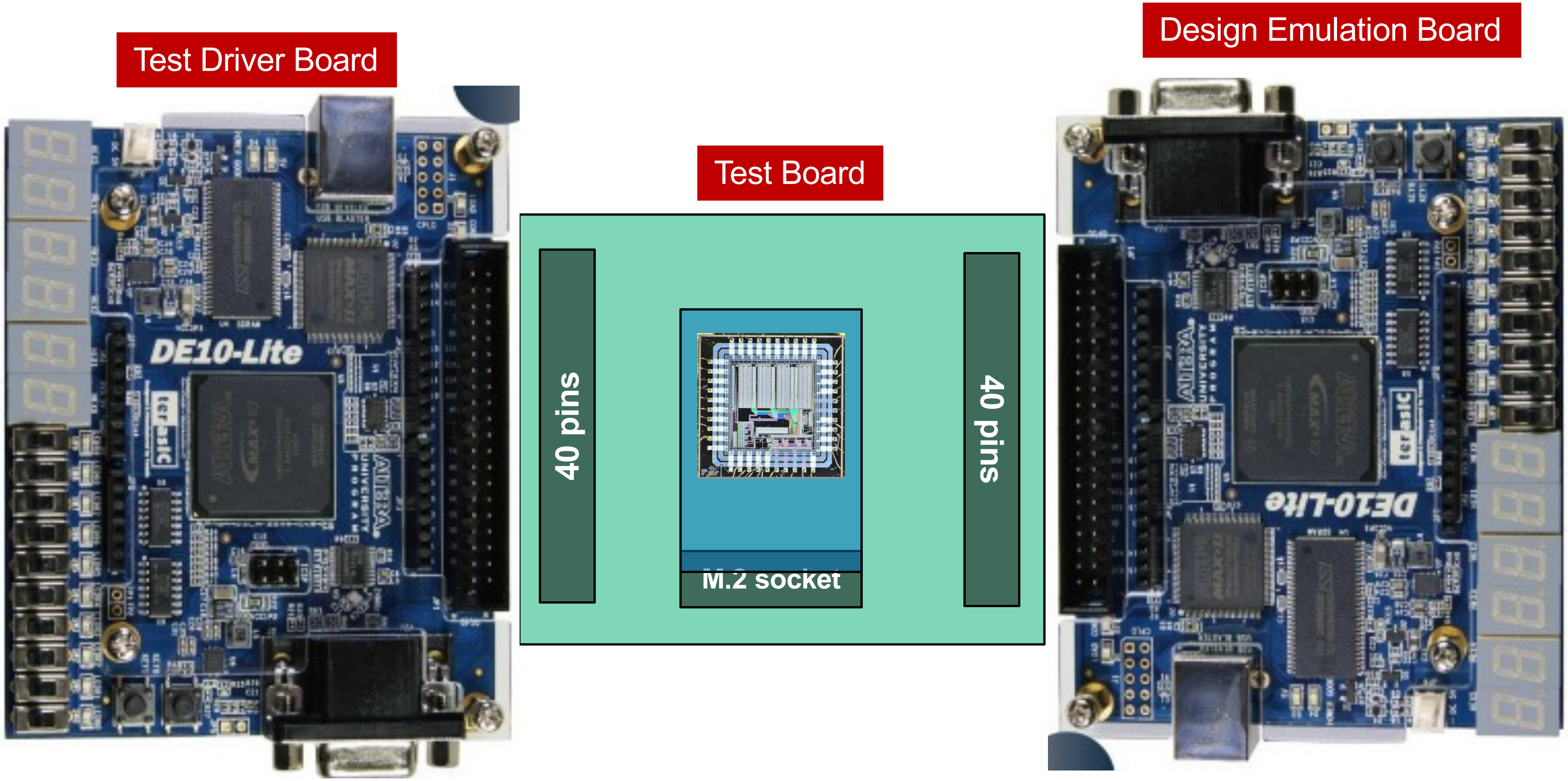
What will be cover today?

- Key dates (Peter C)
- Test Environment (Peter C)
- Design Constraints (Luca)
- Integration Guidelines (Luca)
- Phase Lock Loop and Clocking issues (Peter L)
- Separate group discussion with academic advisors (all 4 staff)

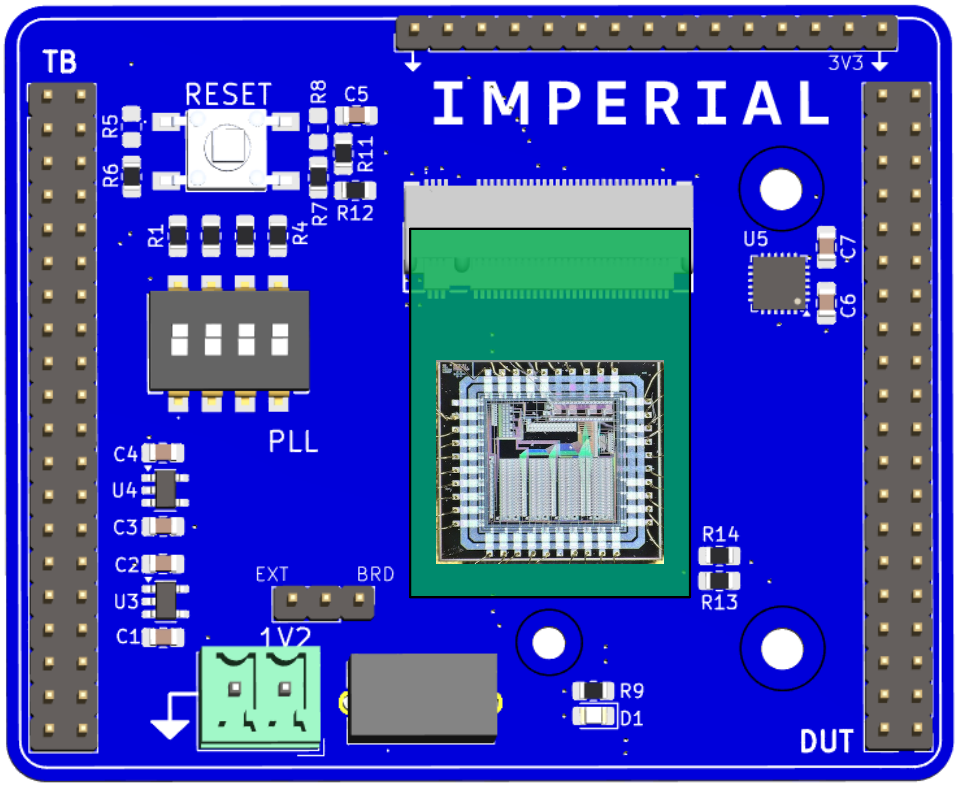
Important Dates

- **5 November** – **Official start** of team chip design projects
- **7 November** – Create and submit Project Github repos
  - Add pykc, luca, peter L and Bougnais to repo as contributors
  - Send Peter team repo URL via email
- **9 November** – Submit Project Specification via Github
- **10 November** – Peter sends 8 project specifications to Apple to recruit engineers as industrial advisors
- **13 Nobember** – Peter releases marking criteria for Team Projects
- **17 November** – Deadline to book TSMC 65nm fab run on 19 February
- **20 November** – Progress review with academic and industrial advisors
- **4 January** – 23.59 nominal deadline for Team Project via Github

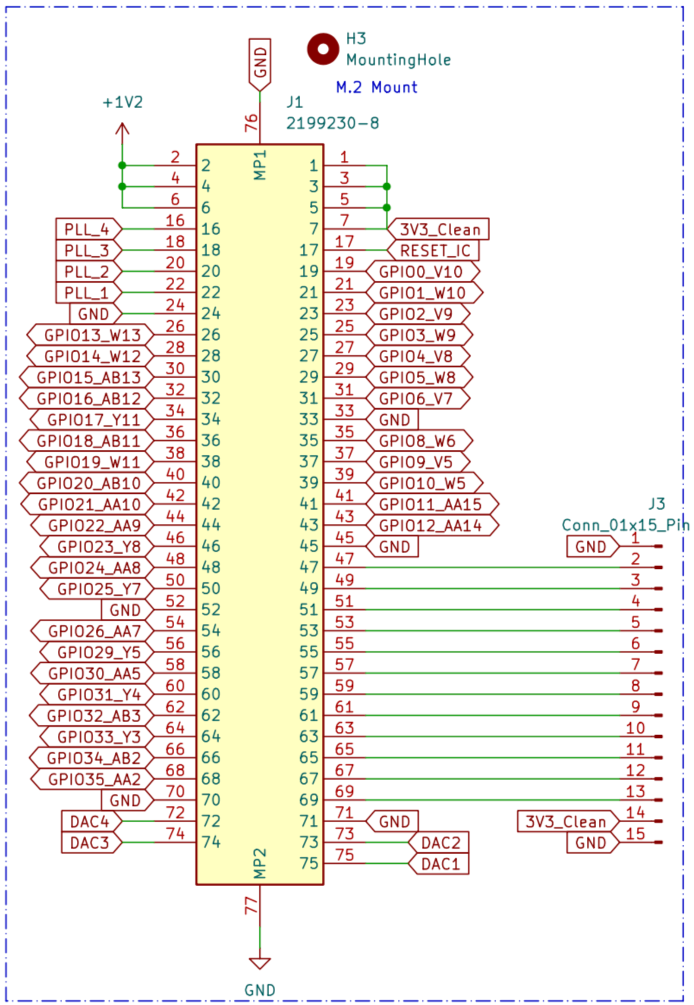
Test Environment



Test Board



M.2 Connector



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Design Constraints and Integration Guidelines

ELEC70142 – Digital VLSI Design

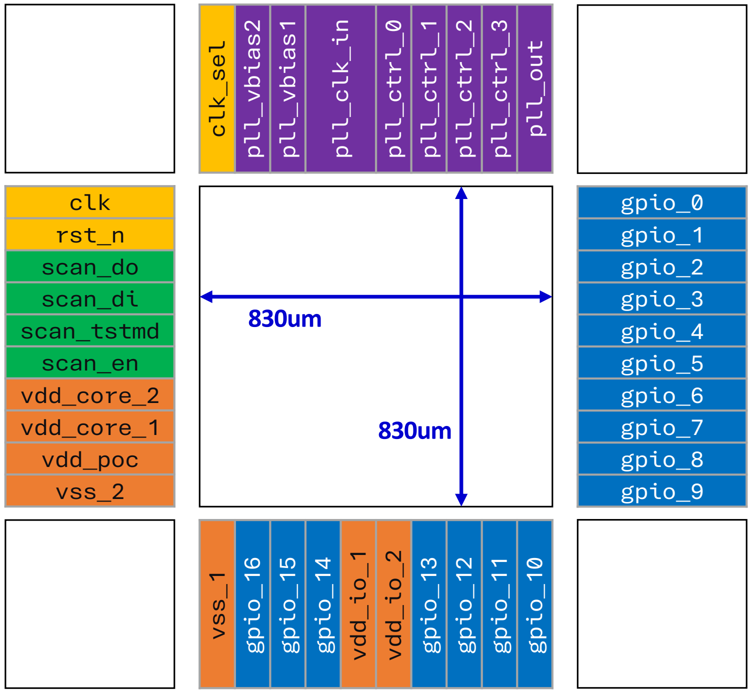
Design Constraints

Area: 830 um x 830 um

Clock Frequency: From 25 MHz to 400 MHz

Input / Outputs: 17 Configurable Digital IOs

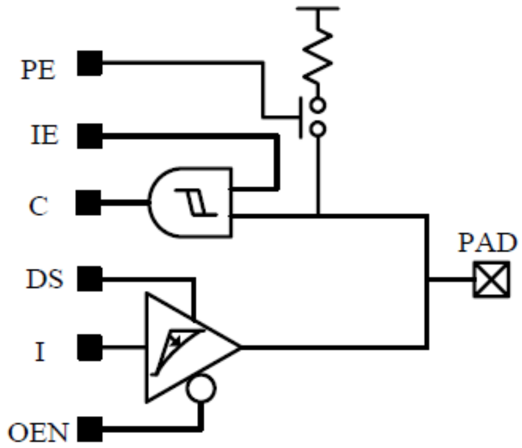
Technology: TSMC 65 – digital libraries used during the labs



Input / Output

- Digital IO cell: PRUW0408SCDG
- Each digital IO cell provides:
  - Input and output enable controls
  - Schmitt Trigger Input
  - Output drive strength for
  - Pull-up resistor

- More information and truth tables in the documentation



Pin Name	Direction	Description
PE	Input	Pull-up enable
IE	Input	Input enable
C	Input	Input signal
DS	Input	Driving Strength
I	Output	Output signal
OEN	Input	Output Enable

Resources
IO Cells Datasheet
/usr/local/cadence/kits/tsmc/beLibs/65nm/TSMCHOME/digital/Documentation/documents/tphn65lpnv2od3_sl_200b/DB_TPHN65LPNV2OD3_SL_TC.pdf

IO Application Note
/usr/local/cadence/kits/tsmc/65n_LP/doc/LIBS/TSMC Slim IO General Application Note_1_.pdf

Verification & Testing

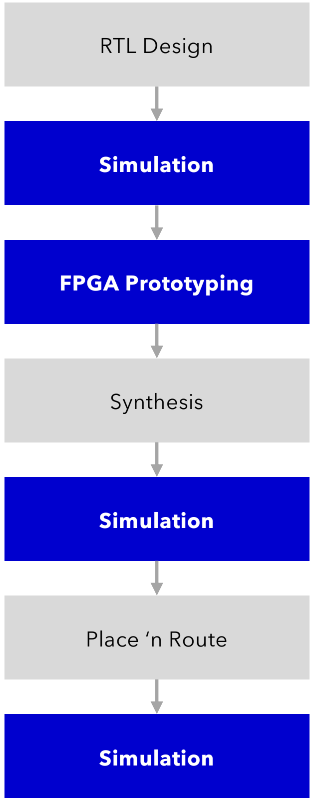
Before Synthesis

- Simulation
  - Verilator (C++)
  - Cocotb (Python)
  - Vivado/Quartus (System Verilog)
  - Xcelium (System Verilog)
- Testing
  - Prototype on DE10 Lite

Post Synthesis/Layout

- Xcelium (SystemVerilog)

Verification takes time!



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Communication Protocols & Peripherals

Communication Interfaces

for loading data, debugging, or connecting peripherals.

- **UART** – Data exchange or debugging with a PC or .
- **SPI** – for sensors or memory devices.
- **I2C** –for low-speed configuration or sensors.

Available Peripherals on DE10-Lite Board

- Switches & Buttons.
- LEDs & 7-Segment Displays.
- Accelerometer (I^2C interface)
- SDRAM (64 MB) – *requires custom interface*.
- VGA Output – *consider number of pins required*.

Resources

DE10 Lite manual

https://ftp.intel.com/Public/Pub/fpgaup/pub/Intel_Material/Boards/DE10-Lite/DE10_Lite_User_Manual.pdf

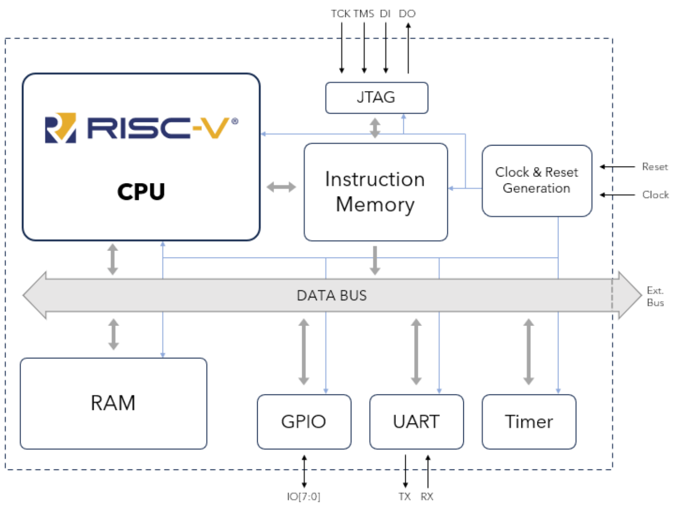
Compute Options

You can add a CPU to your project.
The web offers several Open-Source CPU Cores and SoC:

- Yosis PicoRV32
- ETH PULP Platform
- CV32E40P

… or a simpler SoC written in SystemVerilog implementing a RV32IM CPU with memory-mapped GPIO, UART and Timer peripherals.

- https://github.com/ridoluc/rvcpu-rv32im-soc



Memory Options

On-Chip

- **SRAM (from compiler)**
  - *Preferred option* for implementing larger memories.
  - Typically used for instruction and data memory, buffers, caches, lookup tables.
- **Flip-Flops (Synthesized Memory)**
  - *Use for small memories or register files.*
  - *Synthesized from standard cells — larger area.*
- **Hardcoded (as ROM)**
  - *For small lookup tables, coefficients, or initialization values.*
  - *Implemented from HDL, synthesized using Tie-High/Low cells.*

Off-Chip (DE10-Lite Board)

- **SDRAM**
  - *64MB SDRAM External memory available.*
  - *Requires custom interface due to limited IOs.*
- ***FPGA Flip-Flops***
  - *For simulating external memory*

On-Board Memory Initialization

During Simulation

- Use Verilog initial blocks or `$readmemh` / `$readmemb` to preload memory.
- Convenient for instruction or weight initialization during functional verification.

In the Design

- **Option 1 – JTAG Memory Interface**
  - Custom controller allowing *read/write access* via JTAG instructions.
  - Useful for loading instruction memory or model weights at runtime.
  - Reference implementation and testbenches: github.com/ridoluc/jtag-memory-controller

- **Option 2 – Custom Data Loader**
  - Implement a simple protocol for loading data through **UART**, **SPI**, or other I/O.
  - More effort, but flexible and project-dependent.

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Phase-Locked Loop

Shuanghua Liu
05/11/2025

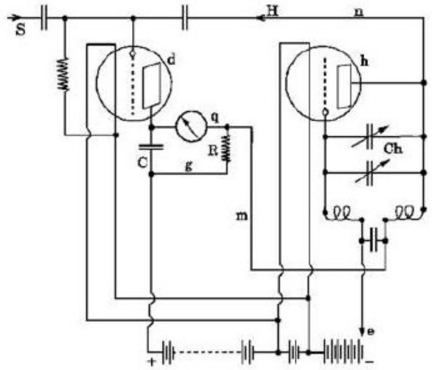
Phase-Locked Loop
What is PLL

• **Phase-Locked Loop (PLL):** A feedback system that compares the **phase difference** between input and output signals.

• Uses an external **reference signal** to control the internal oscillator's **frequency and phase** for synchronization.

• First proposed by **Henri de Bellescize** (France, **1932**).

• Widely adopted since the **1950s**, first in **TV synchronization**, later in **frequency synthesizers** and **radio receivers**.



Phase-Locked Loop

Common use of PLL

Chapter 7 The Oscillator Slim I/O

The oscillator *Slim I/O* cells are designed to oscillate with crystal samples from 2MHz to 30MHz in the fundamental mode, but not designed to work in the KHz band or in overtone oscillation.

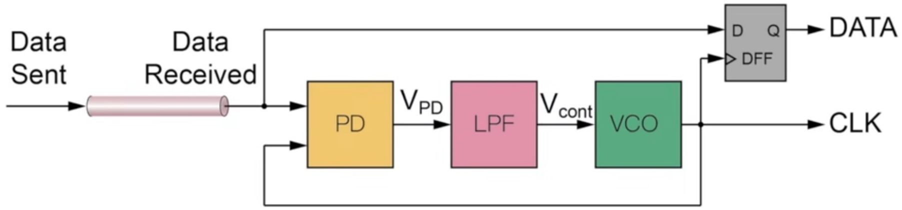
Table 7.1 Selection Guide for the PDXOExDG_G (x =1, 2, 3)

Target Freq (Hz)	2M~ 3M	3M~ 6M	6M~ 10M	10M~ 20M	20M~ 30M
CL (pF)	25	20	16	12	8
Maximum ESR (Ohm)	1K	400	100	80	40
Osc Slim I/O	(I)	(I)	(I)	(I) (II)	(I)~(III)

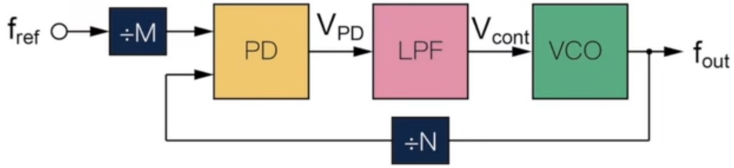
(I) PDXOE1DG_G
(II) PDXOE2DG_G
(III) PDXOE3DG_G

Table 7.2 Selection Guide for the PXOExCDG (x =1, 2)

Target Freq (Hz)	2M~ 6M	6M~ 10M	10M~ 20M	20M~ 30M
CL (pF)	20	16	12	8
Maximum ESR (Ohm)	1K	160	90	40
Osc Slim I/O	(I) (II)	(I) (II)	(I) (II)	(I) (II)



(a) Clock Recovery



(b) Frequency Synthesis

$$f_{out} = \frac{N}{M} f_{ref}$$

Phase-Locked Loop

Phase frequency detector with charge pump

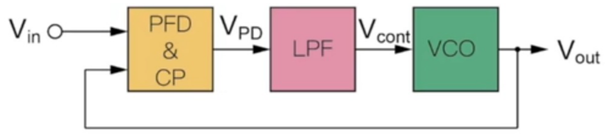
- Produces a highly stable and accurate clock signal based on a crystal oscillator reference

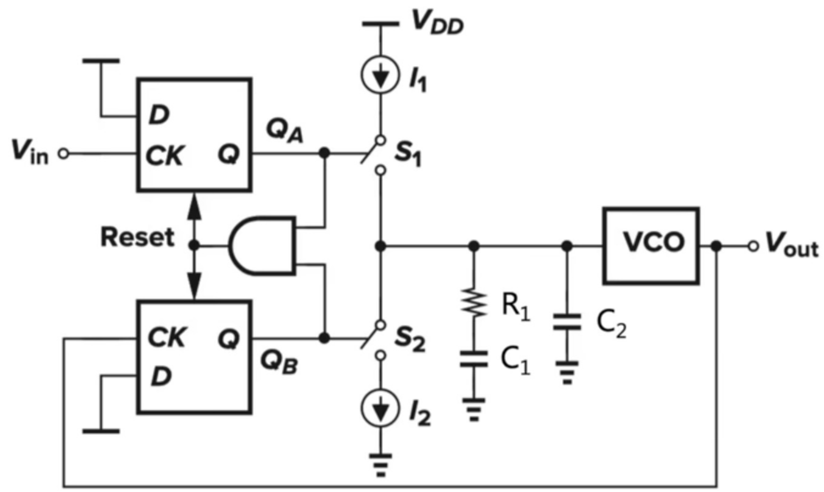
$$
\begin{cases}
w_p = w_{GBW}\dfrac{1+\tan\left(\dfrac{\varphi_{PM\_max}}{2}\right)}{1-\tan\left(\dfrac{\varphi_{PM\_max}}{2}\right)} = w_{GBW}\dfrac{\cos\left(\varphi_{PM\_max}\right)}{1-\sin\left(\varphi_{PM\_max}\right)} \\
w_z = \dfrac{w_{GBW}^2}{w_p}
\end{cases}
$$

ω_{GBW} (radians)	φ_{PM_Max} (degrees)	N	ω_p (radians)	ω_z (radians)
15.7×10^6	60	$1-15$	58.6×10^6	4.3×10^6

$$
\begin{cases}
C_2 = \dfrac{1}{N}\dfrac{I_p}{2\pi}\dfrac{K_{VCO}}{w_{GBW}^2}\sqrt{\dfrac{w_z}{w_p}} \\
C_1 = \left(\dfrac{w_p}{w_z}-1\right)C_2 = \mathbf{13.63}\boldsymbol{C_2} \\
R_1 = \dfrac{1}{C_1 w_z}
\end{cases}
$$

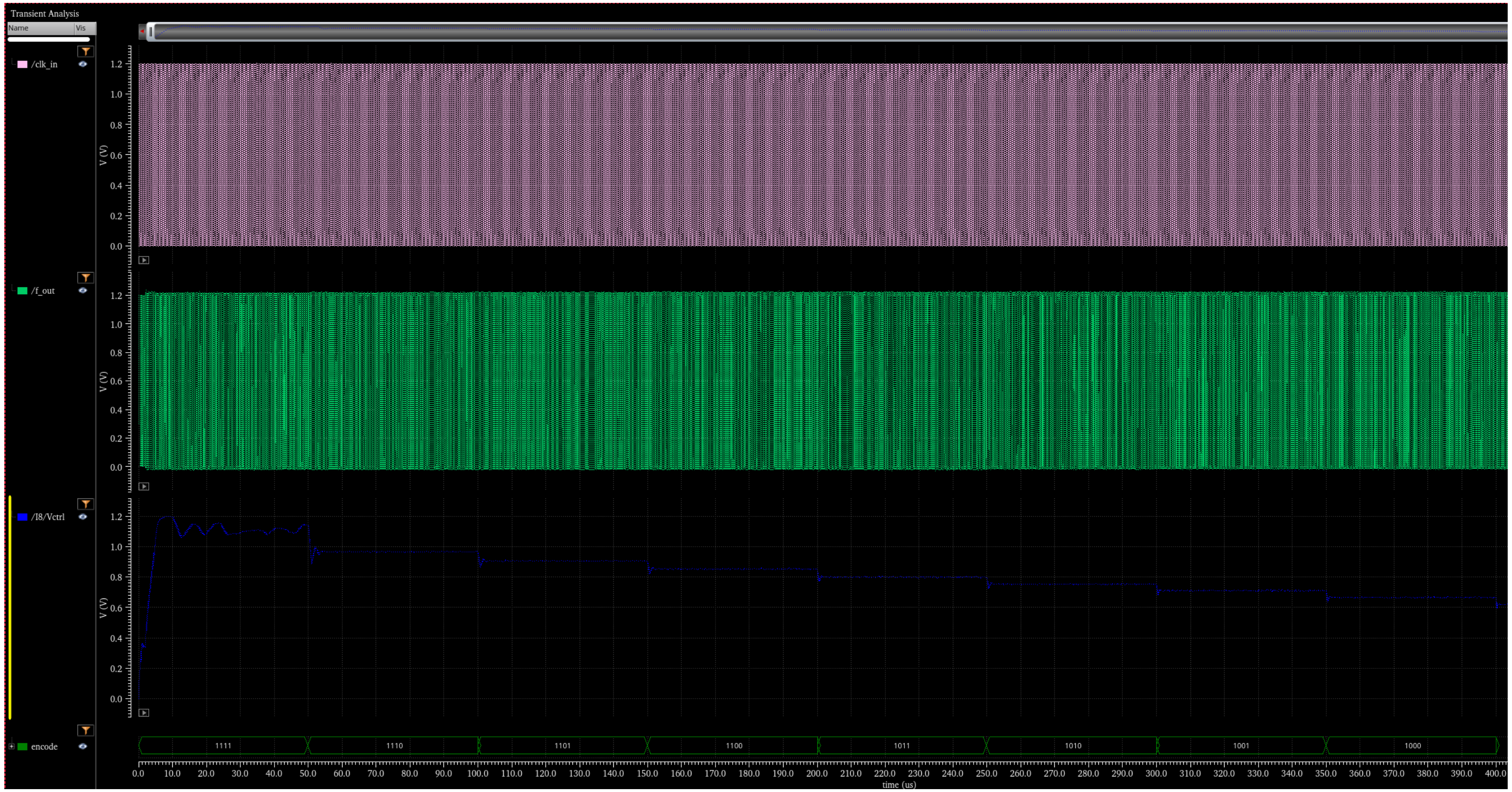
RC_1 (ns)	C_1 (pF)	R (kΩ)	C_2 (pF)	I_{pump} (uA)	K_{VCO} (radians/s·V)
$3.9k$	$59.6p$	36.3	$4.37p$	20	0.5×10^9





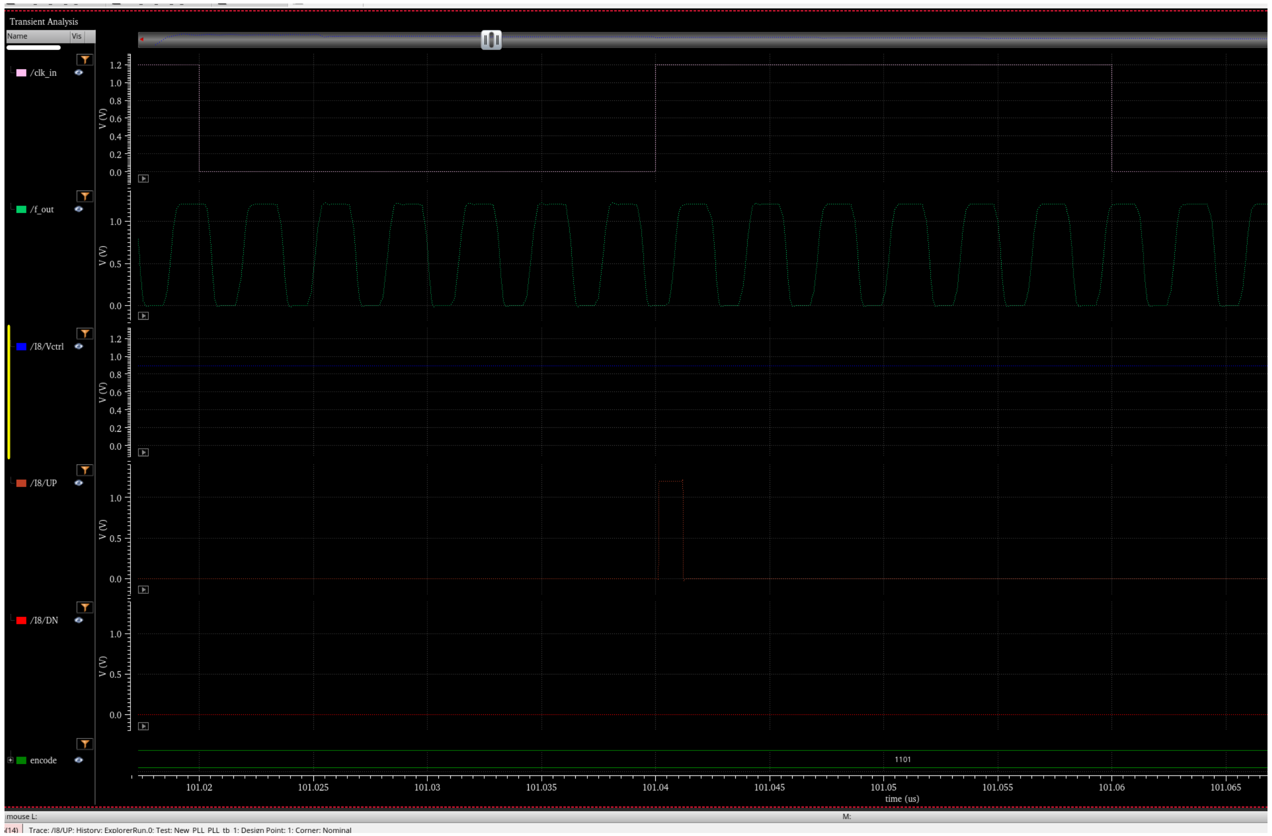
Phase-Locked Loop

Control Voltage adjustment



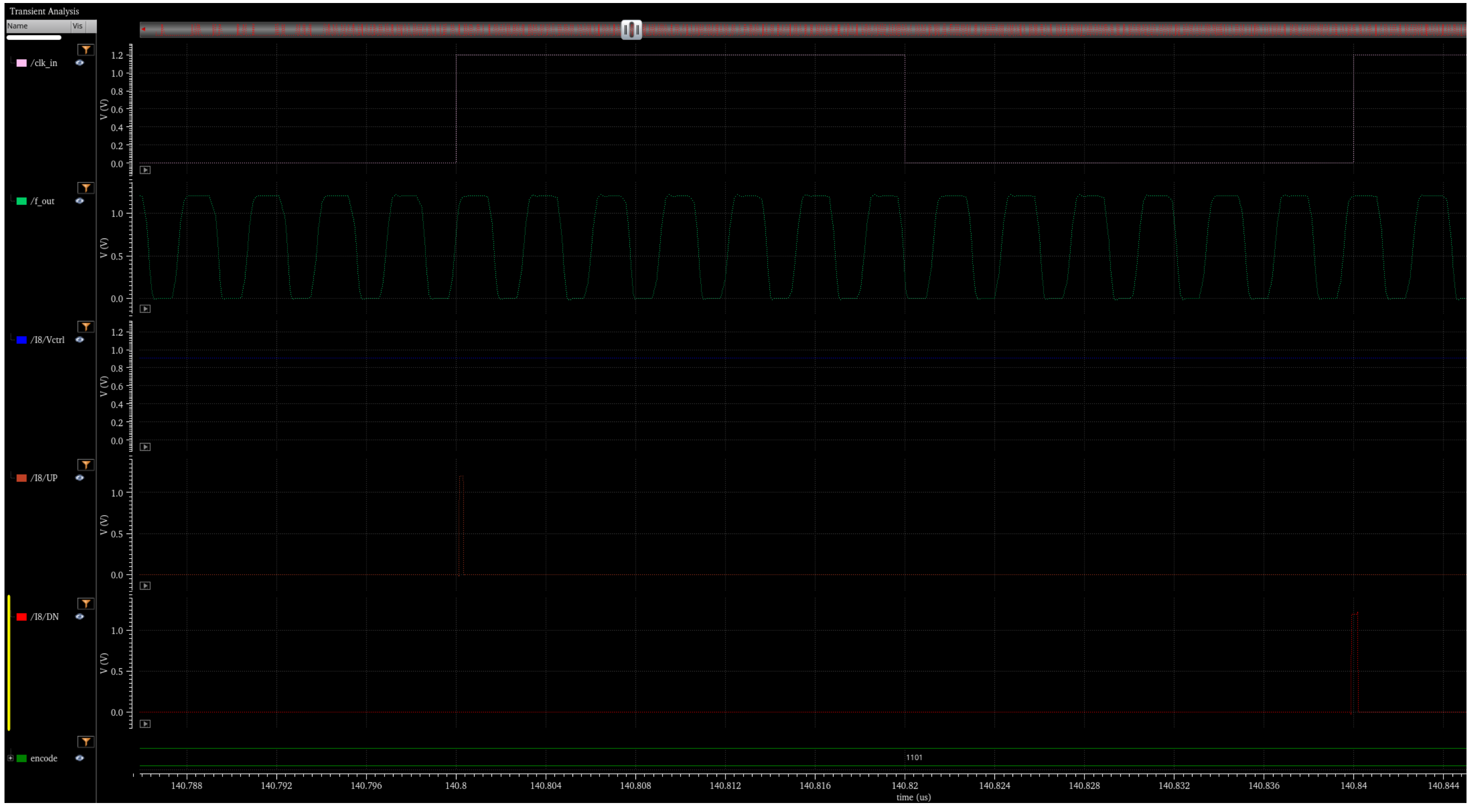
Phase-Locked Loop

Clock mismatch



Phase-Locked Loop

Clock Chasing



Phase-Locked Loop
Takeaway Statistics

- 🧩 **Area:** 300 µm × 195 µm
- ⚡ **Supply Voltage:** 1.2 V
- ⏱️ **Input Clock:** 25 MHz
- 🔄 **Output Clock Range:** 25 – 350 MHz
- 🎛️ **Control Bits:** 4
- 🔋 **Power Consumption:** 3.2 mW

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Thank you

Phase-Locked Loop
05/11/2025